مستشفى الملك فهد
King Fahad Hospital
تجمع الباحة الصحي

PEDIATRIC INTENSIVE CARE UNIT

Policy Name : Assessing Level of Consciousness using Glasgow Coma Scale

Policy Number : ALOCUGCS – PICU 02/3 (9)

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PURPOSE:

1. To evaluate the neurological status
2. To assess intracranial integrity
3. To detect a life threatening complication
4. To record objectively the patient's response to the environment in three major areas: eye opening, movement and verbalization

SCOPE:

First step: Obtaining patient history
Last step: Interpretation of GCS result

ABBREVIATIONS:

GCS - Glasgow Coma Scale

IPSG - International Patient Safety Goals

DEFINITION:

1. **Glasgow Coma Scale** - is a neurological scale that aims to give a reliable, objective way of recording the conscious state of a person for initial as well as subsequent assessment. A patient is assessed against the criteria of the scale.

	Adult	Child	Infant	Score
Eye opening	• Spontaneous • To speech • To pain only • No response	• Spontaneous • To speech • To pain only • No response	• Spontaneous • To speech • To pain only • No response	4 3 2 1
Best verbal response	• Oriented, appropriate • Confused • Inappropriate words • Incomprehensible sounds • No response	• Oriented, appropriate • Confused • Inappropriate words • Incomprehensible sounds • No response	• Coos and babbles • Irritable cries • Cries to pain • Moans to pain • No response	5 4 3 2 1
Best motor response	• Obeys • Localizes • Withdraws	• Obeys commands • Localizes painful stimulus • Withdraws in response to pain • Flexion in response to pain	• Moves spontaneously and purposefully • Withdraws to touch • Withdraws to response in pain	6 5 4

	• Abnormal flexion (decorticate) • Extensor response (decerebrate) • None	• Extension in response to pain • No response	• Abnormal flexion posture to pain • Abnormal extension posture to pain • No response	3 2 1

2. **Head injury is categorized** into 3 levels based on GCS score:
   2.1. Mild head injury: GCS 13-15
   2.2. Moderate head injury: GCS 9-12
   2.3. Severe head injury: GCS 3-8
3. **Terms to explain level of consciousness**
   3.1. **Awake** - may sleep more than usual or be somewhat confused on first awakening, but fully oriented when aroused.
   3.2. **Lethargic** - Drowsy but follows simple commands when stimulated.
   3.3. **Obtunded** - arousable with stimulation; responds verbally with a word or two; follows simple commands; otherwise drowsy.
   3.4. **Stuporous** - very hard to arouse; inconsistently may follow simple commands or speak single words or short phrases; limited spontaneous movement.
   3.5. **Semicomatose** - movements are purposeful when stimulated; does not follow commands or speak coherently.
   3.6. **Comatose** - may respond with reflexive posturing when stimulated or may have no response to any stimulus.

MATERIAL / EQUIPMENT:

1. Torch
2. Hammer

POLICY STATEMENT:

1. Critical care nurse must be familiar with the Glasgow Coma Scale.
2. All patient must be assess for neurological level on admission, per shift and as recommended by the physician.
3. All trauma cases must be assess neurologically every 2-4 hours even without physician's order.
4. Family history should be considered to assess possible genetic disorder with neurologic manifestation.
5. When performing a neurological assessment, the critical care nurse must focus on mental status, cranial nerve functioning and motor status.

6. Expected verbal and motor response must be related to the child's age.

PROCEDURE:

S No	Procedure	Rationale
Preparatory phase:		
1.	Identify the patient by using two identifiers (Name and medical record number – first identification)	Correct identification of patient as per IPSG goal 1
2.	Ensure that the procedure and desired outcome was explained to patient's family, if present	Help to allay anxiety and get family cooperation
3.	Gather material and equipment for the procedure	For easy accessibility
4.	Complete the initial assessment by checking vital signs	Helps to inform the physician, if there is hyperpyrexia or any other abnormal assessment
5.	Obtain family history from parents regarding possible genetic disorder with neurologic manifestation.	A history provides valuable clues regarding causes of neurological deterioration.
Performance phase:		
6.	Assess patient's level of consciousness as follows: 6.1. Awake 6.2. Lethargic 6.3. Obtunded 6.4. Stuporous 6.5. Semi comatose 6.6. Comatose	To provide a baseline of improvement and deterioration in neurologic status.
7.	Examine body surface for the presence of injury, needle marks, petechiea, bites and ticks.	The skin may offer clues to the cause of unconsciousness. Evidence of toxic substances maybe found on the hands, face, mouth and clothing especially in small children.

8.	Assess for pupil size and reactivity.	**Pinpoint** pupils are observed in Poisoning; widely **dilated and reactive** pupils are seen after seizure: **Dilated** pupils maybe due to eye trauma; **Dilated and non reactive** pupils are seen in hypothermia, anoxia, and ischemia.
9.	Assess for motor function: Observation of spontaneous activity, posture, and response to painful stimuli. 9.1. **Localizing Pain-** An appropriate response is to reach up and above shoulder level toward the stimulus. 9.2. **Withdrawal-** An appropriate response is to pull the extremity or body away from the stimulus. 9.3. **Decorticate posture –** adduction of the arms at the shoulder, the arms being flexed on the chest with the wrists flexed and the hands fisted, and the lower extremities being extended and abducted. 9.4. **Decerebrate posture** - characterized by rigid extension and pronation of the arms and legs 9.5. **Flaccid-** No motor response in any extremity.	To provide clues to the location and extent of cerebral dysfunction. A focal motor deficit such as hemiplegia may prevent a bilateral response Seen with severe dysfunction of cerebral cortex. Signs of dysfunction at the level of the midbrain.
10.	Examine cranial nerve reflex.	To assess for brain stem dysfunction.
11.	Check patient's best verbal response. Use Glasgow Coma Scale for documentation.	To evaluate patient's level of consciousness.
Follow up phase:		
12.	**Documents:** 12.1. Documents the findings in flow sheet	Documentation ensures accountability

	nurse's progress note and nursing care plan 12.2. Document nursing assessment and intervention rendered to patient. 12.3. Records any changes in neurologic status	and continuity of care and also to maintain legal record
13.	Monitor any changes in neurologic status and report changes as indicated.	A change in GCS of 2 or more points may be significant and requires investigation. If patient demonstrate deterioration, notify physician without delay to provide immediate intervention.

DOCUMENTATION:

1. PICU Flow chart (FM 052)
2. OASIS system
3. Pediatric Nursing Assessment form

RESPONSIBILITIES:

Physicians are responsible to assess patient and order medications.

Head Nurse responsible to evaluate staff nurses about their knowledge in assessing patient's level of consciousness using the Glasgow Coma Scale.

Charge Nurses are responsible to monitor proper assessment of GCS score.

Staff nurses are responsible to carry out orders, assess GCS score and re-assess patient in a timely manner and to report any signs of deterioration or improvement.

MEASUREMENT:

1. 100% of the staff should know how to use Glasgow Coma scale
2. 100% of newly admitted and discharged patients must have a GCS scale

REFERENCES:

1. Ministry of Health Policy and Procedure, KSA
2. PALS Provider Manual 2006-2007
3. Critical Care Nursing: A Holistic Approach Lippincott and Williams 8th Edition

DISTRIBUTION:

1. Nursing Administration
2. Quality Department
3. Office of the Executive Director General

REVIEW SUMMARY:

1. Addition of policy statement